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Solution:

the number of pulses: $\text{Floor}[3+(n/3)]$

Here, the last digit of my ID is =3

So the number of pulses = 3

Now, the maximum DC power dissipation **without simulations** based on the total thermal resistance and the maximum allowed temperatures is:

Total thermal resistance $\theta_{JA,total} = R1 + R2 + R3 + R4 = 4.5 + 7.4 + 10.9 + 56.6 = 79.4^\circ\text{C}/\text{W}$

Max Temperature difference $(\Delta T_{max}) = T_{junction,max} - T_{ambient} = 125^\circ - 20^\circ = 105^\circ$

So, minimum DC power $(P_{DC,max}) = \Delta T_{max} / \theta_{JA,total} = 105^\circ - 79.4^\circ\text{C}/\text{W} = 1.322 \text{ W}$

Now, here are the three things affecting power dissipation in Power MOSFET.
(N.B.: I have written it with pen and paper, please do consider this)

The three thing affecting power dissipation in Power MOSFET

1. On-state resistance ($R_{DS(on)}$) :-

This is the channel resistance when the MOSFET is fully ON and it is the conduction loss due to the load current (I_D) :-

$$P_{\text{conduction-loss}} \approx I_{D, \text{rms}}^2 \cdot R_{DS(on)}$$

2. Switching frequency :- (f_s)

The rate at which transistor switches ON/OFF

$$P_{\text{sw, loss}} = E_{\text{sw}} \cdot f_s$$

(E_{sw} = switching loss energy)

3. Total Gate charge (Q_g) :-

The total charge required to switch the device. This parameter is directly influence by the gate driver loss and indirectly to the switching loss energy (E_{sw}).

$$P_{\text{gate}} \approx Q_g \cdot V_{\text{gs}} \cdot f_s$$

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Fig 1: Parameter affecting Power dissipation in Power MOSFET

Now, after putting the parameters and selecting the time period = 10ms, the maximum power dissipation is 178.74W (That value comes from the electrical circuit to thermal circuit parameter shifting, where the voltage value in the thermal circuit is considered as power dissipation)

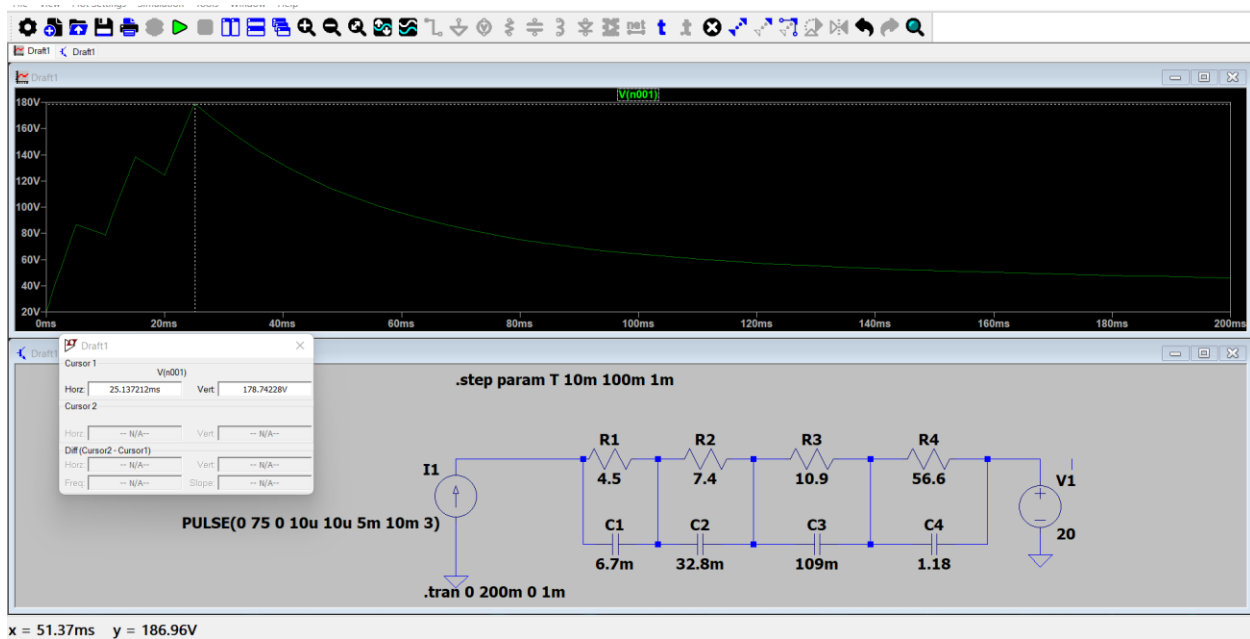


Fig 2: Power dissipation after selecting the Time Period of 10ms

Observation: Here we can see that if we lower the time period, it will increase the time frequency like: $f=1/T$. that means the $T_{junction}$ will increase because the heat from one pulse hasn't dissipated completely before the next pulse hits.

***Now, if we increase the time period from 10ms to 100ms, the curve becomes:

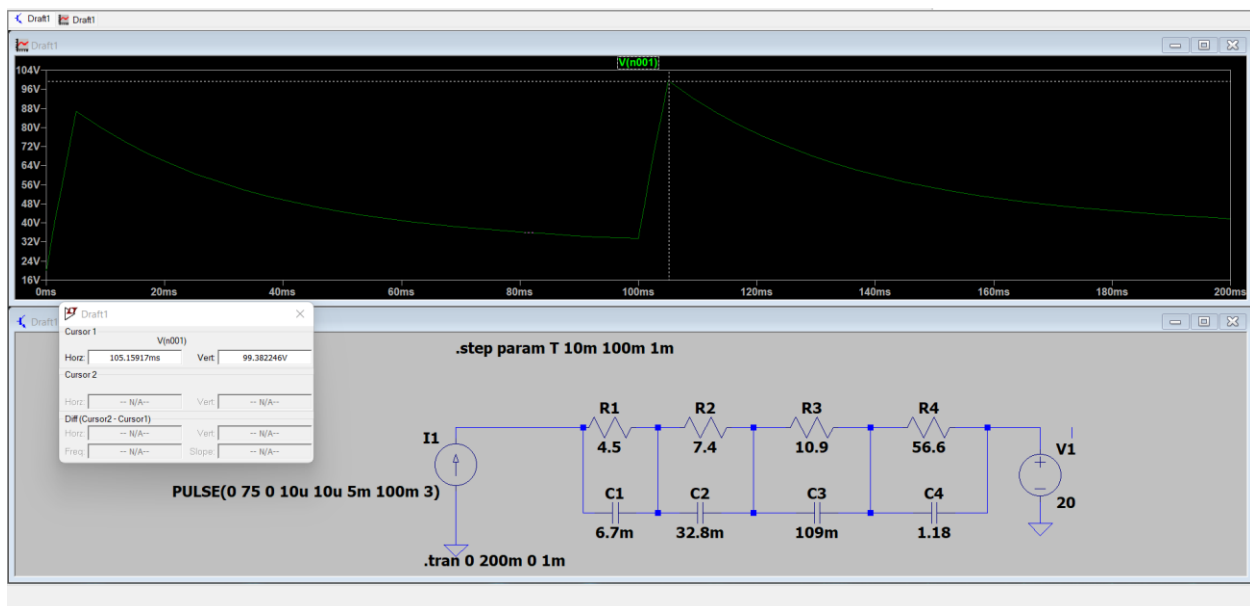


Fig 3: Power dissipation after selecting Time Period of 100ms

Observation: Here we can see that if we increase the time period, it will decrease the time frequency like: $f=1/T$. that means the $T_{junction}$ will decrease because the MOSFET has more time to cool down before the next pulse arrives.

That is why the Power dissipation decreases from 178.7W to 99.3W

Now, I will vary the time period from 10m to 100m with an increase of 1m, and the result is shown in Fig.4

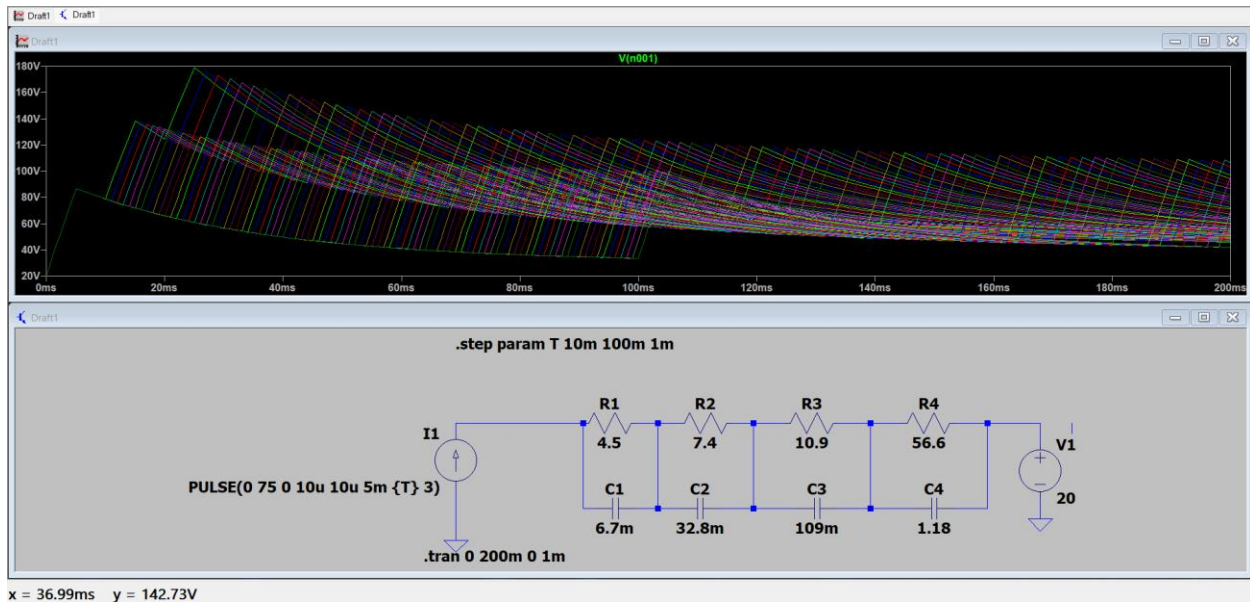


Fig 4: The power dissipation curve with varying time period from 10ms to 100ms with an increasing of 1ms.

The maximum pulsing frequency is shown in Fig .5, so that the MOSFET does not break down (max T junction 125 °C) in room temperature (20 °C) conditions after a pulse burst of floor 3 pulses.

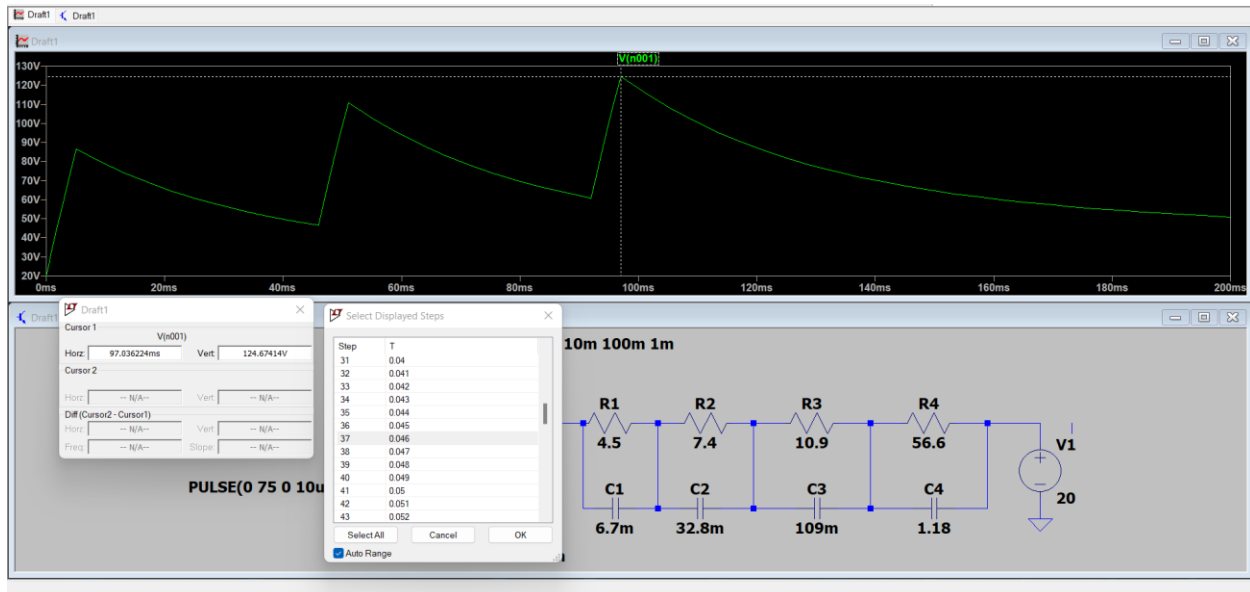


Fig 5: Maximum Pulsing Frequency that MOSFET does not break at maximum junction temperature of 125°.

